

The colour of Earth's sky, 4.4 billion years to today

What the sky looked like from the ground—straight up and at the horizon, at the equator and the poles, at noon and at dusk—at thirteen moments in Earth's history, reconstructed with a spectral radiative-transfer model.

Prepared with Claude (Anthropic) · September 2026

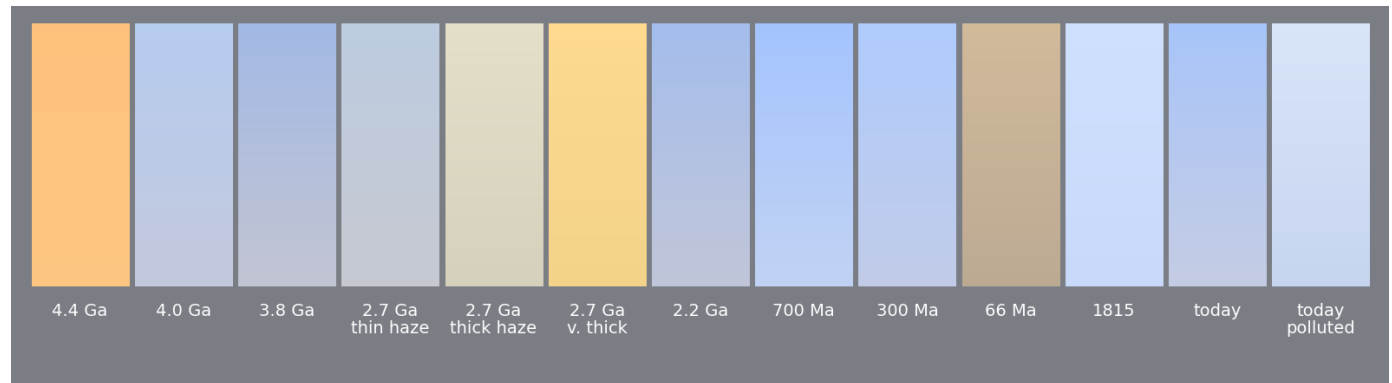


Figure 1. Mid-latitude noon sky at thirteen epochs, zenith (top) to horizon (bottom), oldest at left. Brightness shown relative to today's clean sky.

Five things the sky did

1. Under the 30-bar CO_2 atmosphere of the early Hadean the sky was a shadow-less peach-white glow with no visible Sun.
2. During the hazy Neoproterozoic the whole dome was cream to apricot, with almost no difference between zenith and horizon.
3. Before the Great Oxidation Event there was no ozone, so every twilight ended with a cream-coloured, not blue, zenith. A blue dusk is an optical fingerprint of oxygen.
4. Snowball Earth had the bluest sky in Earth's history; the K–Pg impact winter had no sunsets at all.
5. The zenith-to-horizon colour gradient is the single best diagnostic of an atmosphere's state: largest in clean air, gone under haze or soot, reversed under volcanic sulfate.

Has this been done before?

Not as such. A search of the Valency corpus (arXiv, PubMed, Earth-ArXiv and others) finds the two halves of the problem solved separately but never joined. On one side, atmospheric-optics work computes modern sky colour rigorously: Lynch and Mazur (2005) derive CIE chromaticities of the sky, Sun and distant objects from MODTRAN spectra; Peyvandi et al. (2016) simulate 5.6 million daylight spectra with SBDART to map how aerosol and ozone shift sky colour; Lee and Mollner (2017) show with Monte Carlo simulation that vivid solar-horizon twilight colours need aerosol while the purest antisolar colours occur in a purely molecular atmosphere. On the other side, paleoatmosphere work constrains composition: Catling and Zahnle (2020) for the Archean (O_2 below 10^{-6} of present, CO_2 10–2,500 $\times$ modern, CH_4 10^2 – $10^4\times$ modern), and Arney et al. (2016), Mak et al. (2023) and Goldblatt et al. (2024) for the organic-haze episodes. Arney's "Pale Orange Dot" computes what a hazy Archean Earth looks like *from space*, as an exoplanet spectrum, and that is as close as the literature comes. Nobody has computed the view from the ground, or the horizon-versus-zenith, latitude, and sunset structure. So this report does.

How the colours were computed

For each epoch the atmosphere is built from components with their own vertical profile and spectral optical depth: Rayleigh-scattering gas (N_2 , O_2 , CO_2 , CH_4 , H_2O in their epoch-appropriate partial pressures), ozone (Chappuis band), tropospheric aerosol, a stratospheric sulfate layer, a tholin-like fractal organic haze (strongly blue-absorbing, forward-scattering), and impact soot. Sky radiance is integrated numerically along each line of sight through a spherical-shell atmosphere, so grazing sunlight at twilight is handled properly, with multiple

scattering and surface reflection added from a delta-Eddington two-stream solution. The young Sun is treated as a blackbody made fainter and slightly cooler following standard solar evolution: 5,560 K and 70% of today's luminosity at 4.4 Ga, 5,620 K and 75% at 3.8 Ga, 5,660 K and 80% at 2.7 Ga, 5,700 K and 85% at 2.2 Ga, 5,772 K today. The colour effect is only ~ 200 K, barely perceptible and far smaller than the atmospheric effects; the brightness effect is what makes the early skies dimmer. Spectra from 380–780 nm are converted to CIE XYZ and shown as sRGB. The model reproduces measured modern values (clear zenith 10,000–20,000 K, horizon 6,000–8,500 K, setting Sun $\sim 1,700$ K, pink antisolar Belt of Venus) before being pointed backward.

The young Sun. The Sun is treated as a blackbody whose effective temperature and luminosity follow standard solar-evolution tracks: 5,560 K and 70% of today's output at 4.4 Ga, rising through 5,620 K (75%) at 3.8 Ga, 5,660 K (80%) at 2.7 Ga and 5,700 K (85%) at 2.2 Ga to 5,772 K today (Table 1). The colour effect is small: a 5,560 K Sun is only about 200 K cooler in colour temperature than the modern one, a difference that is barely perceptible side by side and is swamped by atmospheric effects ten to a hundred times larger. The brightness effect is not small, and is why the early-epoch skies are rendered dimmer. Fraunhofer lines and the young Sun's ultraviolet excess, which matter for haze photochemistry but not for visible colour, are ignored.

The model is provided as the Python source `skymodel.py`; `run_epochs.py` produces the sky colours and `limb_grid.py` the limb and disk colours for the renderings. Epoch atmospheres are summarised in Table 1.

Findings

Horizon versus zenith

In a clean molecular atmosphere the zenith is blue and the horizon white because the horizon path is long enough that blue is scattered back out before reaching the eye and multiple scattering fills in the other colours. The size of that gradient is the clearest single diagnostic of atmospheric state. It is largest in the cleanest atmospheres (Snowball Earth, today's clean air: 8,000–13,000 K of colour-temperature difference), shrinks as Rayleigh depth rises (late Hadean), collapses to nothing when a broadband absorber or thick scatterer sits overhead (hazy Archean, impact winter, 30-bar Hadean), and actually *reverses* under a stratospheric sulfate veil or heavy pollution, where the zenith becomes whiter than the horizon.

Table 1: Atmospheres used for each epoch. Gas amounts are partial pressures in bar; AOD is tropospheric aerosol optical depth at 550 nm; haze, sulfate and soot are column optical depths at 550 nm; T_{eff} and $L_{\odot}$ are the assumed solar effective temperature and luminosity relative to today.

Epoch	Gas (bar)	O ₃ (DU)	AOD	Haze	Sulfate	Soot	T_{eff} (K)	$L_{\odot}$
4.4 Ga Hadean	CO ₂ 30, N ₂ 1, H ₂ O 0.3	0	0.30	–	–	–	5560	0.70
4.0 Ga Hadean	N ₂ 1, CO ₂ 0.5	0	0.15	–	–	–	5600	0.72
3.8 Ga Archean	N ₂ 0.8, CO ₂ 0.1, CH ₄ 0.001	0	0.08	–	–	–	5620	0.75
2.7 Ga thin haze	N ₂ 0.8, CO ₂ 0.05, CH ₄ 0.003	0	0.08	0.15	–	–	5660	0.80
2.7 Ga thick haze	N ₂ 0.8, CO ₂ 0.05, CH ₄ 0.005	0	0.08	0.6	–	–	5660	0.80
2.7 Ga very thick haze	N ₂ 0.8, CO ₂ 0.05, CH ₄ 0.01	0	0.08	1.5	–	–	5660	0.80
2.2 Ga post-GOE	N ₂ 0.8, O ₂ 0.002, CO ₂ 0.02	150	0.10	–	–	–	5700	0.85
700 Ma Snowball	N ₂ 0.78, O ₂ 0.02, CO ₂ 0.01 (+dust 0.05)	250	0.02	–	–	–	5750	0.94
300 Ma Carboniferous	N ₂ 0.78, O ₂ 0.33	330	0.15	–	–	–	5765	0.975
66 Ma impact winter	modern air	300	0.15	–	0.5	1.5	5772	0.995
1815 volcanic year	modern air	300	0.10	–	0.4	–	5772	1.00
Modern, clean	N ₂ 0.78, O ₂ 0.21, Ar 0.01	300	0.10	–	–	–	5772	1.00
Modern, polluted	modern air	300	0.60	–	–	–	5772	1.00

Table 2: Summary of sky colours by epoch (mid-latitude noon unless stated).

Epoch	Noon zenith	Horizon	Sunset
4.4 Ga, 30-bar CO ₂	Peach-white, 3,300 K, shadowless	Same as zenith	Whole dome dims to orange; Sun rarely visible
4.0 Ga, 0.5-bar CO ₂	Pale milky blue, 10,500 K	Milk white	Deep red Sun; cream twilight (no ozone)
3.8 Ga, clear Archean	Blue, 13,000 K	White	Modern-like, then yellow-beige twilight zenith
2.7 Ga, thick haze	Cream-tan, 5,600 K	Cream-tan	Vermilion horizon, amber sky, long orange dusk
2.2 Ga, post-GOE	Blue, 13,700 K	White	Blue twilight zenith returns
700 Ma, Snowball	Deep blue, 17,000 K	Blue-white, stays blue	Golden, pale
300 Ma, 33% O ₂	Blue, 13,600 K	White, slightly brighter	Modern-like; smoky more often
66 Ma, impact winter	Dim amber-beige, 4,600 K	Beige	None; dome fades to brown
1815, volcanic year	Milky blue-white, 9,300 K	Slightly bluer than zenith	Salmon Sun, pink-lavender afterglow
Today, clean	Blue, 15,000 K	White, 8,400 K	Orange horizon, pink antisolar

Equator, mid-latitudes, poles

Latitude enters through solar elevation and surface albedo. A high Sun lights the sky from above through a short path, giving a brighter but paler zenith (equatorial noon $\sim 9,000$ K); a low Sun sends light through more air, so the polar summer zenith is a deeper, darker blue ($\sim 19,000$ K today, 20,000 K over Snowball ice) while the polar horizon is warmer and yellower. The same geometry makes the poles the place where any overhead absorber is most visible: the hazy Archean sky is cream at the equator but distinctly orange-yellow at the poles, and the impact-winter Sun that is still a pale disc at the equator is gone at 75° latitude. Snow and ice brighten and blue-shift the whole dome by reflecting light back into the atmosphere, which is why polar and Snowball horizons stay blue where a mid-latitude land horizon goes white.

Particulates

Three particle regimes appear in the record and they look different from one another. Non-absorbing sulfate high in the stratosphere (volcanic years) whitens the day sky and produces long, luminous pink and lavender twilights, because the layer is still sunlit after the ground is dark. Absorbing organic haze (Neoproterozoic) or soot (impact winter) removes blue outright, turning the dome tan or amber with almost no directional structure, and dims the surface by factors of 3–20. Boundary-layer pollution or dust brightens and greys the zenith, and buries the setting Sun in murk before it reaches the horizon. Tropospheric aerosol is also what makes sunset colour vivid: the model confirms Lee & Mollner’s result that a purely molecular sky yields a pale, yellow-orange horizon while any turbidity deepens it to red.

Sunsets and sunrises through time

The biggest surprise is ozone. Today the sky above a setting Sun grades from orange through pale yellow to a zenith that stays blue well into twilight; that persistent blue is not Rayleigh scattering but ozone absorption of yellow and orange light along the grazing path. Before the Great Oxidation Event there was no ozone, so every sunset for the first two billion years ended with the zenith fading through cream and beige to orange, and the sky never turned that deep twilight blue. The return of a blue dusk is, in this model, a direct optical

consequence of oxygenation at ~ 2.4 –2.2 Ga. Beyond that: Hadean sunsets were not events at the horizon but a slow dimming of the whole glowing dome; hazy Archean sunsets were the longest and most saturated in Earth’s history, vermillion at the horizon and amber overhead; Snowball sunsets were brief and golden; the sunsets of the impact winter did not exist as colour at all; and the reddest, most theatrical skies of the Phanerozoic belong to volcanic years like 1815 and 1883.

The atmosphere from space

To show latitude and altitude at once, these renderings look at Earth from space with the Sun behind the viewer at the equinox, so the solar zenith angle at every point equals its latitude (equator at the centre line, poles at top and bottom). The shell around the disk is the atmosphere’s limb: the colour of light scattered toward the viewer along a tangent ray at each altitude, computed with the same model. The atmosphere is drawn 30 times too thick, so 100 km spans half an Earth radius; a real limb is a hairline. The disk shows the planet’s reflected colour over a dark ocean (over ice for Snowball Earth), with limb brightness compressed so faint upper layers remain visible.

Modern, clean air. Modern Earth for reference: white-blue troposphere grading to a deep blue stratosphere and fading to black, brightest at the equator and darker toward the poles.

Early Archean, ~ 3.8 Ga. Clear early Archean. Structurally modern: white-blue near the surface, deep blue above, fading to black. With a little more CO₂ and no ozone the shell is marginally paler than today’s and the disk very slightly whiter.

Neoproterozoic, ~ 2.7 Ga (thick haze). Hazy Neoproterozoic. The organic haze colours the entire limb tan: every tangent ray below ~ 60 km crosses the layer (centred ~ 45 km) twice. Only above it does the shell turn a faint, dusty blue. The disk is the pale orange dot, though from space it is a subtle cream-orange rather than a vivid one.

Early Hadean, ~ 4.4 Ga. 30-bar CO₂ Hadean. The shell is an opaque, luminous white-peach almost all the way up, because tangent rays at any tangent height below ~ 60 km pass through many optical depths of CO₂; only the outermost fringe thins to blue. The disk is bright and nearly white: a cloudless, Rayleigh-only Venus.

Snowball Earth, ~ 700 Ma. Snowball Earth. The bluest shell in the set above a blazing white disk of ice. The polar limb stays bright because the frozen surface reflects so much light back into the atmosphere.

Carboniferous, ~ 300 Ma. Carboniferous, 33% O₂. The extra oxygen raises total pressure by about a tenth, so the limb is marginally thicker and brighter than today’s, but to the eye it is indistinguishable from modern Earth: once ozone and clean air are in place, the sky from space has been the same pale blue dot for two billion years.

K-Pg impact winter, 66 Ma. Impact winter. A dim brown-grey band from the soot and

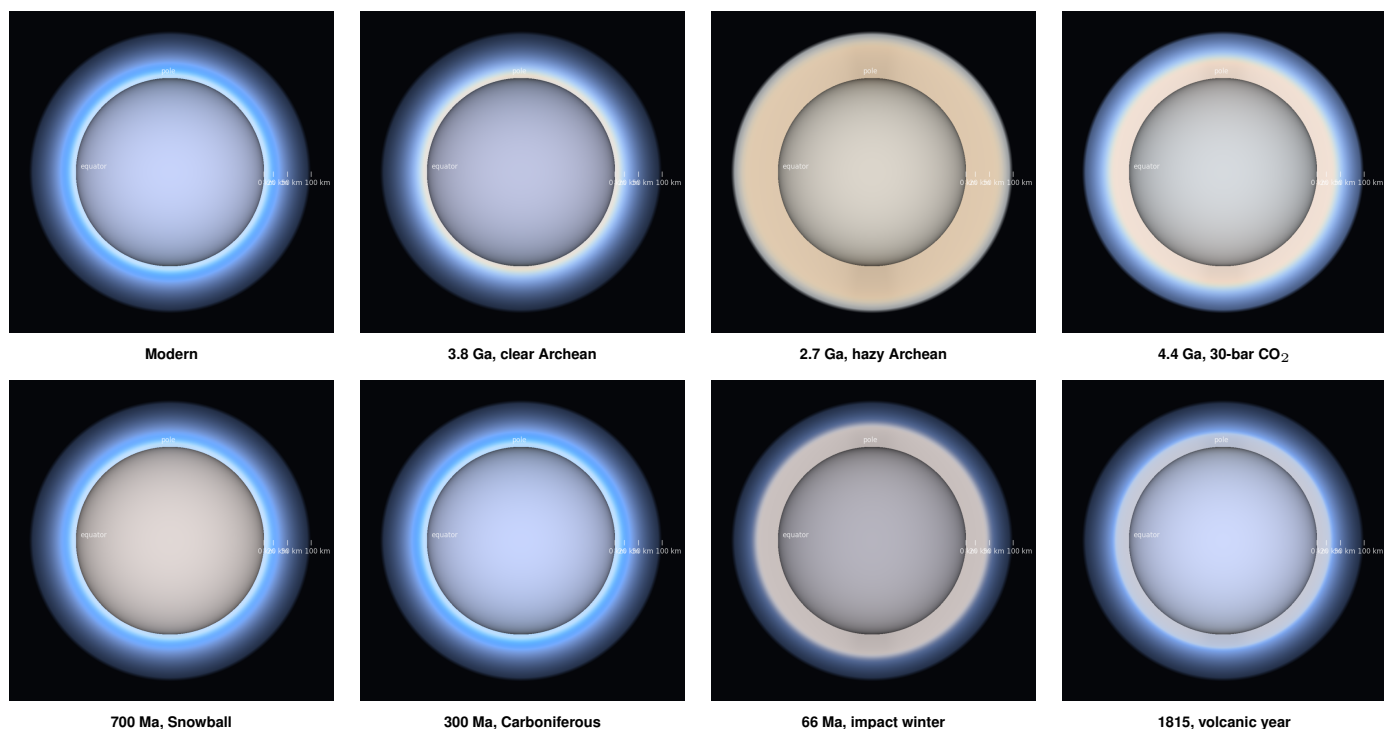


Figure 2: Volumetric renderings of the atmosphere at eight epochs. Equinox geometry with the Sun behind the viewer; the shell is the limb colour along tangent rays, drawn 30 times too thick (100 km spans half an Earth radius); the disk is the planet's reflected colour over ocean (ice for Snowball Earth). Limb brightness is compressed so faint upper layers remain visible.

sulfate layer occupies the lower and middle shell; above ~40 km the untouched upper atmosphere is still faintly blue. The disk is a dull grey-lilac, and dark at high latitudes.

Volcanic year (Tambora 1815 / Toba-lite). Volcanic year. A milky white sulfate band near 20 km sits between the white-blue troposphere and the deep blue upper shell. The disk is brighter and whiter than clean modern Earth.

Epoch by epoch

The thirteen panels that follow are laid out identically so they can be compared at a glance. Each panel begins with the epoch's name and a one-line summary of the atmosphere assumed (Table 1 gives the full parameters). The upper row shows the noon sky dome at three latitudes—the equator, a mid-latitude site and the summer pole—with the zenith at the top of each swatch and the horizon at the bottom, the correlated colour temperatures of both printed above, and the Sun's disc drawn in its own colour, at its noon elevation, and with a brightness that reflects how much of it survives the atmosphere (where the Sun would not be visible at all, the swatch says so). The two strips beneath trace the sky from the solar horizon across the zenith to the antisolar horizon at two moments: with the Sun sitting on the horizon, and with it 4° below, in civil twilight. All swatches are shown at a brightness relative to today's clean sky, so a dim epoch reads as dim; the paragraph after each panel explains what the colours mean and why they arise.

Early Hadean, ~4.4 Ga

post-magma-ocean CO₂/steam atmosphere (~30 bar CO₂)

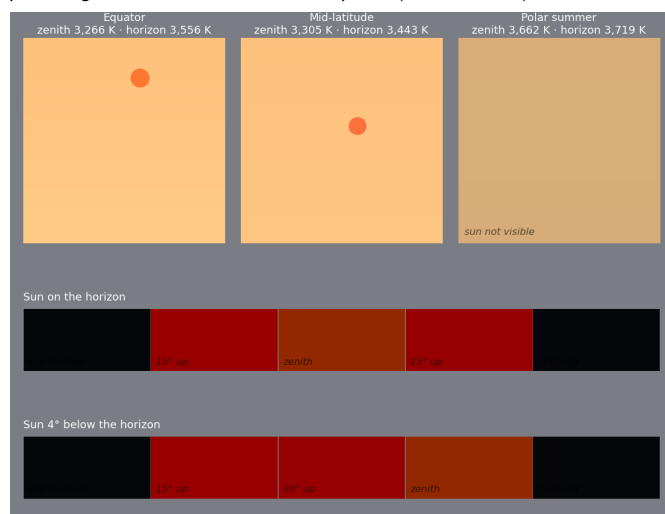


Figure 3: Early Hadean, ~4.4 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A ~30-bar CO₂ atmosphere has a Rayleigh optical depth near 5 at 550 nm and about 18 in the violet, so blue light is scattered back to space many times before it reaches the ground. What arrives is a diffuse, shadowless, peach-white glow of ~3,300 K, nearly identical from zenith to horizon, under a young Sun that is only 70% as bright. The Sun itself is a dim ember at the equator and vanishes entirely at mid and high latitudes: this is a Venus-like sky. Sunset is a brief, deep-orange dimming of the whole dome rather than an event at the horizon.

Late Hadean, ~4.0 Ga

CO_2 mostly drawn down into carbonate; ~ 1 bar N_2 , ~ 0.5 bar CO_2

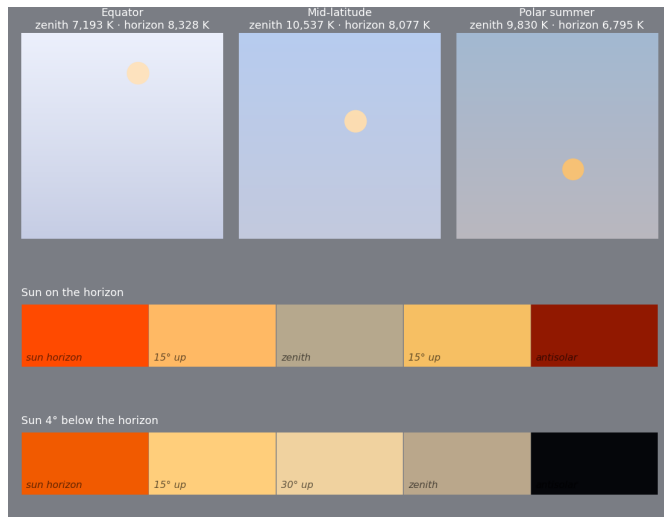


Figure 4: Late Hadean, ~4.0 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Once most of the CO_2 has gone into carbonate, Rayleigh scattering falls to roughly twice the modern value. The sky turns pale blue at the zenith but conspicuously whiter than today (7,000–10,500 K versus 9,000–15,000 K), because the extra scattering fills the dome with multiply-scattered white light. Horizons are milky. The setting Sun is deep red and the twilight zenith, with no ozone yet, is cream rather than blue.

Early Archean, ~3.8 Ga

haze-free, anoxic; $\text{CO}_2 \sim 0.1$ bar, $\text{CH}_4 \sim 1000$ ppm, no ozone

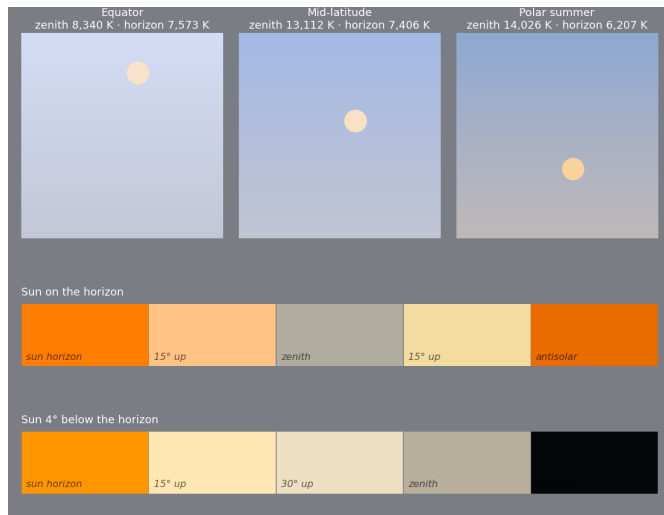


Figure 5: Early Archean, ~3.8 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

With CO_2 down to ~ 0.1 bar, noon skies are close to modern blue, only slightly whiter. The unmistakable difference is at dusk. The blue of the modern twilight zenith is produced by ozone absorbing yellow-orange light (the Chappuis band) along the long grazing path; with no free O_2 there is no ozone, so the Archean twilight zenith is a pale yellow-cream, and the whole sky fades from blue to beige to orange as the Sun sets. Sunsets themselves are much like today's, slightly redder from the extra CO_2 and dimmer Sun.

Neoproterozoic, ~2.7 Ga (thin haze)

biogenic $\text{CH}_4/\text{CO}_2 \sim 0.1$: intermittent thin organic haze

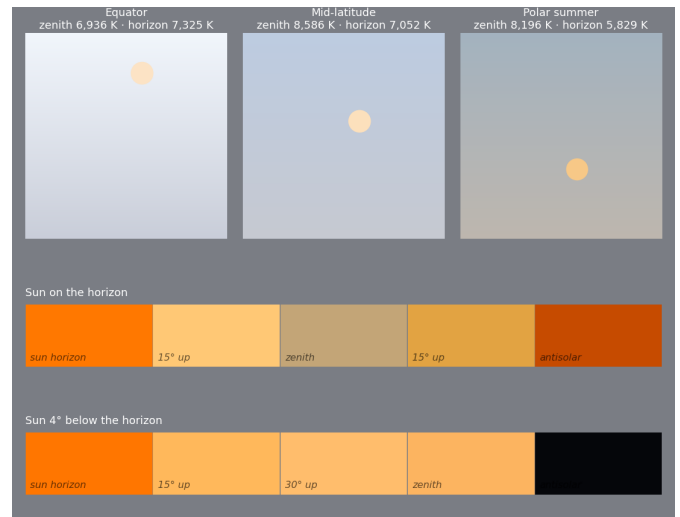


Figure 6: Neoproterozoic, ~2.7 Ga (thin haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

An intermittent organic haze at optical depth ~ 0.15 pushes the zenith toward a pastel, milk-blue (7,000–8,500 K) and lays a warm cast on the horizon; at the poles, where sunlight crosses the haze obliquely, the horizon turns visibly cream. Twilight is orange all the way to the zenith.

Neoproterozoic, ~2.7 Ga (thick haze)

"Pale Orange Dot": Titan-like fractal organic haze, $\tau_{550} \sim 0.6$

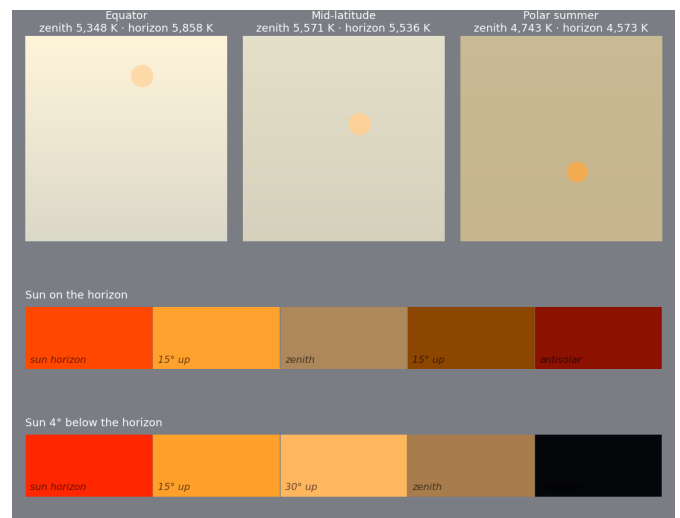


Figure 7: Neoproterozoic, ~2.7 Ga (thick haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

This is the 'Pale Orange Dot' seen from below. At optical depth ~ 0.6 the haze absorbs blue and scatters forward, so the entire dome is a warm cream to pale tan ($\sim 5,300$ K at the equator, $\sim 4,700$ K at the poles) with almost no zenith-to-horizon gradient: the sky looks like a bright overcast made of orange-tinted glass. The Sun is a soft, defined disc (haze scatters strongly forward) at 4,100 K. Sunsets are deep and long: the solar horizon reaches vermilion, the sky 15° above the Sun is amber, and the antisolar sky is peach rather than pink.

Neoproterozoic, ~2.7 Ga (very thick haze)
upper-end haze, $\tau_{550} \sim 1.5$; CH_4/CO_2 well above 0.2

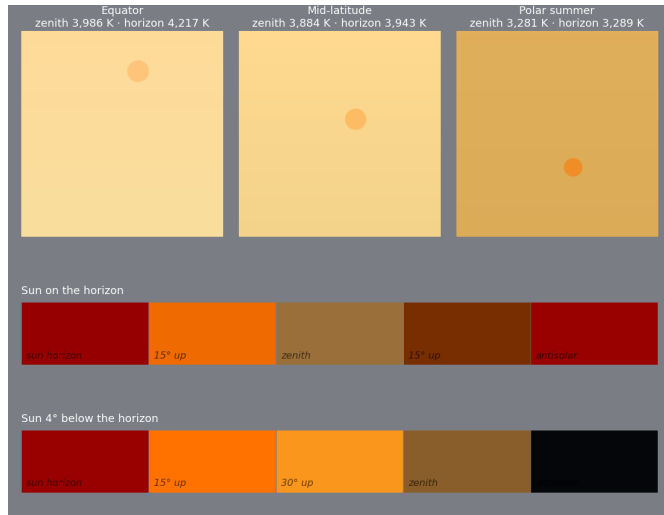


Figure 8: Neoproterozoic, ~2.7 Ga (very thick haze). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

At the upper end of plausible haze ($\tau \approx 1.5$) the sky is a uniform saturated apricot (3,300–4,200 K), fully Titan-like, and the polar Sun is reduced to an orange smear. Anything below this haze would have had roughly 20–30% of the surface sunlight of a clear sky.

Paleoproterozoic, ~2.2 Ga
after the Great Oxidation Event: $\text{O}_2 \sim 1\%$ PAL, first ozone layer (~150 DU)

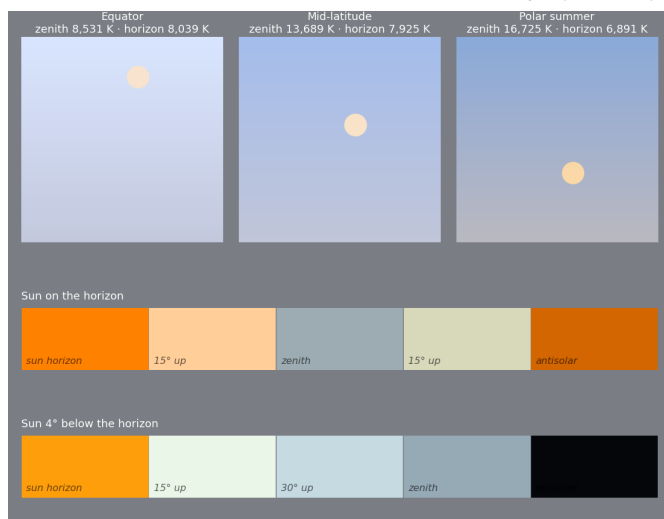


Figure 9: Paleoproterozoic, ~2.2 Ga. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

After the Great Oxidation Event a few percent of today's O_2 builds an ozone column of order 150 DU, about half the modern value. Noon skies are essentially modern blue. Twilight recovers its blue zenith, though a paler one than today's. (The pink antisolar arch, by contrast, needs no ozone: it is a Rayleigh effect and shows up faintly even in the Hadean model.) From here to the present, the daytime sky is blue everywhere; what changes is the aerosol load, the surface albedo, and the Sun's brightness.

Snowball Earth, ~700 Ma
cold, dry, very clean air; ice/snow surface (albedo 0.75); some dust

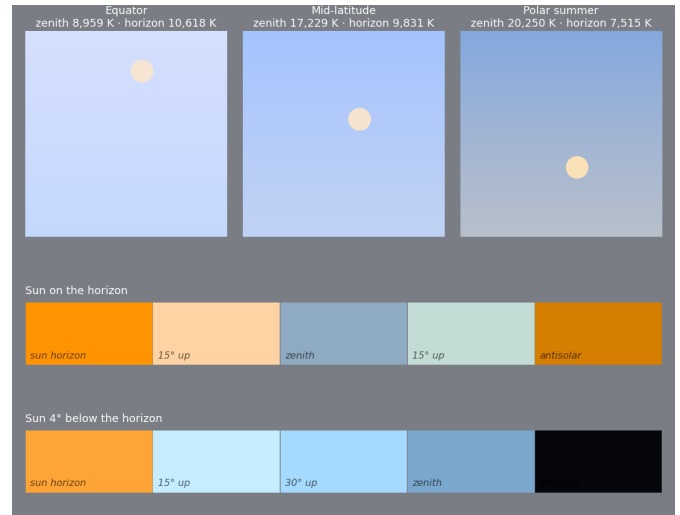


Figure 10: Snowball Earth, ~700 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A frozen planet has a dry, exceptionally clean atmosphere and a surface albedo near 0.75. The result is the bluest sky in Earth's history: 9,000 K at the equatorial zenith and over 20,000 K over polar ice, with horizons that stay distinctly blue instead of whitening, because the sunlit ice floods the sky from below with blue-white light. Sunsets are golden and comparatively pale; there is little aerosol to redden them.

Carboniferous, ~300 Ma
 $\text{O}_2 \sim 30\text{--}35\%$; total pressure ~1.1 bar; wildfire smoke common

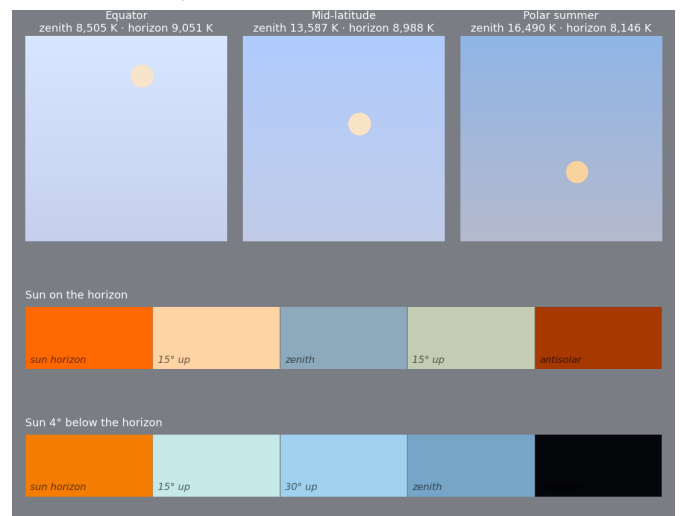


Figure 11: Carboniferous, ~300 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Thirty-plus percent O_2 raises the total pressure and Rayleigh scattering by ~10%, so skies are marginally brighter and horizons a touch whiter than today; the effect is subtle. Frequent wildfire smoke in the high- O_2 world would have produced red suns and brown-orange horizons far more often than in the modern era.

K-Pg impact winter, 66 Ma

months after Chicxulub: stratospheric soot ($\tau \sim 1.5$) + sulfate ($\tau \sim 0.5$)

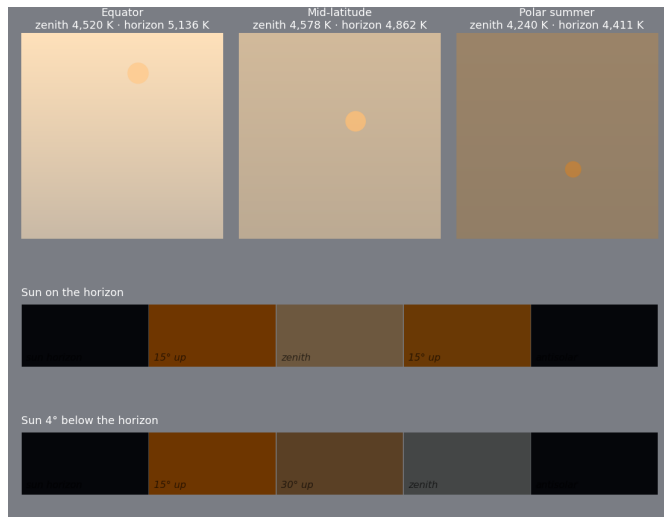


Figure 12: K-Pg impact winter, 66 Ma. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Months after the Chicxulub impact, a global stratospheric layer of soot and sulfate turns the sky into a dim, uniform amber-beige (4,200–5,100 K) with no blue at all, about one-fifth as bright as a clear sky at the equator and 20× dimmer at high latitudes. The Sun is a pale orange disc at low latitudes and effectively gone near the poles. Sunsets do not happen as colour events: the whole dome simply fades to dark brown, and the twilight sky is grey.

Volcanic year (Tambora 1815 / Toba-lite)

stratospheric sulfate veil, $\tau_{550} \sim 0.4$

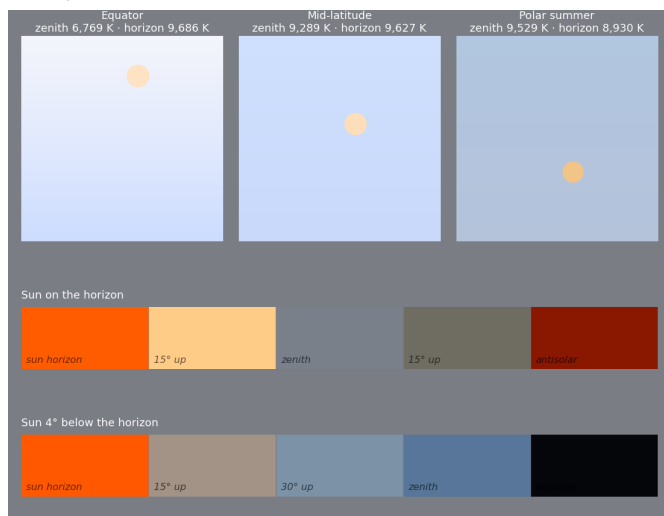


Figure 13: Volcanic year (Tambora 1815 / Toba-lite). Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

A Tambora-scale sulfate veil ($\tau \approx 0.4$) is what most people picture as a spectacular sky, and the model agrees. Noon skies are milky, whiter at the zenith than the horizon (the reverse of a clean sky). Sunsets glow salmon-orange, and the after-sunset sky 15–30° above the horizon turns a luminous pink-lavender: the high sulfate layer is still lit when the lower atmosphere is in shadow. This is the sky of Turner's 1816 paintings and of the Krakatoa twilights of 1883.

Modern, clean air

aerosol optical depth 0.1, ozone 300 DU

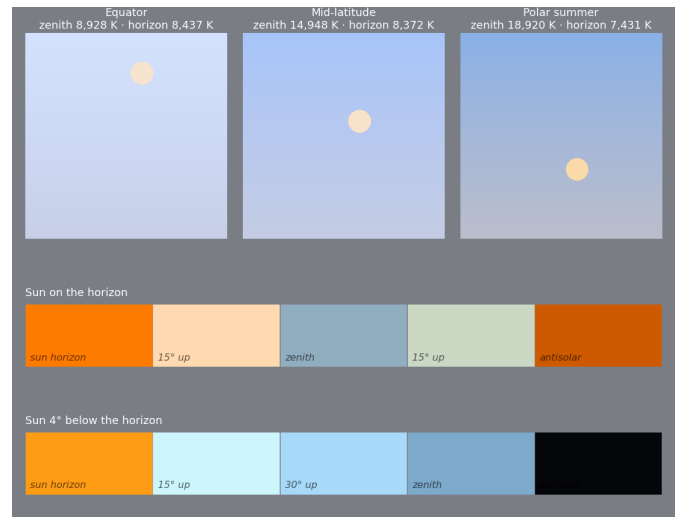


Figure 14: Modern, clean air. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Baseline. Equatorial zenith $\sim 9,000$ K; mid-latitude $\sim 15,000$ K; polar summer $\sim 19,000$ K. Horizons run 7,400–8,400 K, essentially white. Setting Sun $\sim 1,700$ K; solar horizon orange; antisolar horizon pink.

Modern, polluted megacity

aerosol optical depth 0.6, slightly absorbing

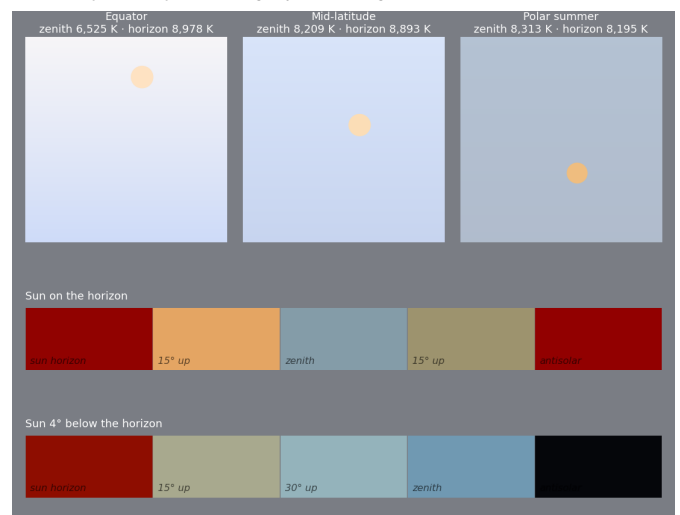


Figure 15: Modern, polluted megacity. Top: noon sky dome at the equator, mid-latitude and polar summer, zenith at top, horizon at bottom, Sun's disc in its own colour. Below: sky from the solar horizon to the antisolar horizon with the Sun on the horizon and 4° below it.

Aerosol optical depth 0.6 flattens the sky to a near-uniform pale grey-blue (6,500–8,300 K), brighter and whiter at the zenith than a clean sky, with the horizon slightly bluer than the zenith. The Sun disappears into grey murk well before it reaches the horizon.

Caveats

Paleoatmospheric compositions carry order-of-magnitude uncertainty, and the haze optical properties are taken from Titan-analog tholins; the three haze cases bracket that uncertainty. Aerosol loads for every pre-Cenozoic epoch are educated guesses. The colour-matching functions use an analytic fit, and colours are shown without chromatic adaptation, so the tints represent what a modern camera set to daylight balance would record rather than what an adapted observer would "see" (an adapted eye would perceive the hazy Archean sky as nearer to white and the modern sky as bluer than shown). Clouds are omitted throughout. The two-stream multiple-scattering term is approximate for the thickest atmospheres, where a full Monte Carlo

treatment would refine the exact tint of the Hadean sky.

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Appendix: computed chromaticities

Table 3: CIE 1931 chromaticity (x, y) and correlated colour temperature of the noon zenith and horizon sky, and direct-Sun brightness relative to the modern equatorial Sun.

Epoch	Latitude	Zenith (x, y)	CCT (K)	Horizon (x, y)	CCT (K)	Sun
Early Hadean, ~4.4 Ga	Equator	0.421, 0.400	3,266	0.406, 0.400	3,556	0.0087
Early Hadean, ~4.4 Ga	Mid-latitude	0.418, 0.399	3,305	0.412, 0.400	3,443	0.0021
Early Hadean, ~4.4 Ga	Polar summer	0.399, 0.391	3,662	0.396, 0.392	3,719	1.4e-06
Late Hadean, ~4.0 Ga	Equator	0.303, 0.317	7,193	0.292, 0.302	8,328	0.63
Late Hadean, ~4.0 Ga	Mid-latitude	0.275, 0.288	10,537	0.294, 0.305	8,077	0.56
Late Hadean, ~4.0 Ga	Polar summer	0.277, 0.296	9,830	0.309, 0.320	6,795	0.25
Early Archean, ~3.8 Ga	Equator	0.291, 0.303	8,340	0.299, 0.310	7,573	0.77
Early Archean, ~3.8 Ga	Mid-latitude	0.264, 0.274	13,112	0.301, 0.313	7,406	0.72
Early Archean, ~3.8 Ga	Polar summer	0.259, 0.272	14,026	0.318, 0.330	6,207	0.47
Neoproterozoic, ~2.7 Ga (thin haze)	Equator	0.306, 0.323	6,936	0.301, 0.316	7,325	0.72
Neoproterozoic, ~2.7 Ga (thin haze)	Mid-latitude	0.288, 0.304	8,586	0.305, 0.320	7,052	0.64
Neoproterozoic, ~2.7 Ga (thin haze)	Polar summer	0.290, 0.312	8,196	0.325, 0.340	5,829	0.31
Neoproterozoic, ~2.7 Ga (thick haze)	Equator	0.336, 0.355	5,348	0.324, 0.346	5,858	0.46
Neoproterozoic, ~2.7 Ga (thick haze)	Mid-latitude	0.331, 0.353	5,571	0.332, 0.353	5,536	0.35
Neoproterozoic, ~2.7 Ga (thick haze)	Polar summer	0.355, 0.377	4,743	0.361, 0.379	4,573	0.07
Neoproterozoic, ~2.7 Ga (very thick haze)	Equator	0.386, 0.396	3,986	0.377, 0.396	4,217	0.19
Neoproterozoic, ~2.7 Ga (very thick haze)	Mid-latitude	0.393, 0.405	3,884	0.391, 0.406	3,943	0.1
Neoproterozoic, ~2.7 Ga (very thick haze)	Polar summer	0.429, 0.422	3,281	0.429, 0.423	3,289	0.004
Paleoproterozoic, ~2.2 Ga	Equator	0.289, 0.302	8,531	0.294, 0.305	8,039	0.87
Paleoproterozoic, ~2.2 Ga	Mid-latitude	0.262, 0.272	13,689	0.295, 0.306	7,925	0.8
Paleoproterozoic, ~2.2 Ga	Polar summer	0.252, 0.264	16,725	0.307, 0.320	6,891	0.5
Snowball Earth, ~700 Ma	Equator	0.287, 0.296	8,959	0.275, 0.286	10,618	0.99
Snowball Earth, ~700 Ma	Mid-latitude	0.253, 0.260	17,229	0.279, 0.292	9,831	0.92
Snowball Earth, ~700 Ma	Polar summer	0.245, 0.256	20,250	0.298, 0.316	7,515	0.61
Carboniferous, ~300 Ma	Equator	0.289, 0.303	8,505	0.286, 0.295	9,051	0.91
Carboniferous, ~300 Ma	Mid-latitude	0.262, 0.273	13,587	0.286, 0.296	8,988	0.82
Carboniferous, ~300 Ma	Polar summer	0.251, 0.266	16,490	0.293, 0.304	8,146	0.42
K-Pg impact winter, 66 Ma	Equator	0.361, 0.368	4,520	0.342, 0.353	5,136	0.12
K-Pg impact winter, 66 Ma	Mid-latitude	0.359, 0.367	4,578	0.350, 0.360	4,862	0.054
K-Pg impact winter, 66 Ma	Polar summer	0.371, 0.374	4,240	0.365, 0.370	4,411	0.00039
Volcanic year (Tambora 1815 / Toba-lite)	Equator	0.309, 0.324	6,769	0.281, 0.292	9,686	0.67
Volcanic year (Tambora 1815 / Toba-lite)	Mid-latitude	0.283, 0.297	9,289	0.281, 0.293	9,627	0.53
Volcanic year (Tambora 1815 / Toba-lite)	Polar summer	0.280, 0.298	9,529	0.285, 0.301	8,930	0.13
Modern, clean air	Equator	0.286, 0.299	8,928	0.291, 0.301	8,437	1
Modern, clean air	Mid-latitude	0.258, 0.267	14,948	0.291, 0.302	8,372	0.92
Modern, clean air	Polar summer	0.247, 0.259	18,920	0.300, 0.313	7,431	0.54
Modern, polluted megacity	Equator	0.313, 0.327	6,525	0.286, 0.298	8,978	0.6
Modern, polluted megacity	Mid-latitude	0.292, 0.306	8,209	0.286, 0.299	8,893	0.46
Modern, polluted megacity	Polar summer	0.290, 0.308	8,313	0.291, 0.308	8,195	0.082